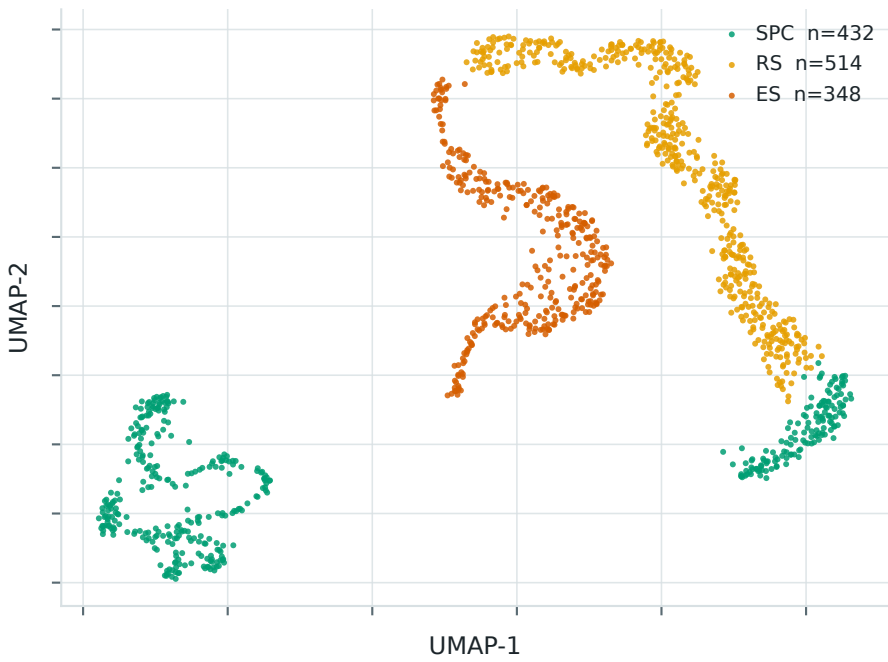


Spermatogenesis positive control: PeakATail recovers progressive 3'UTR shortening only for PAS it places in real 3'UTRs

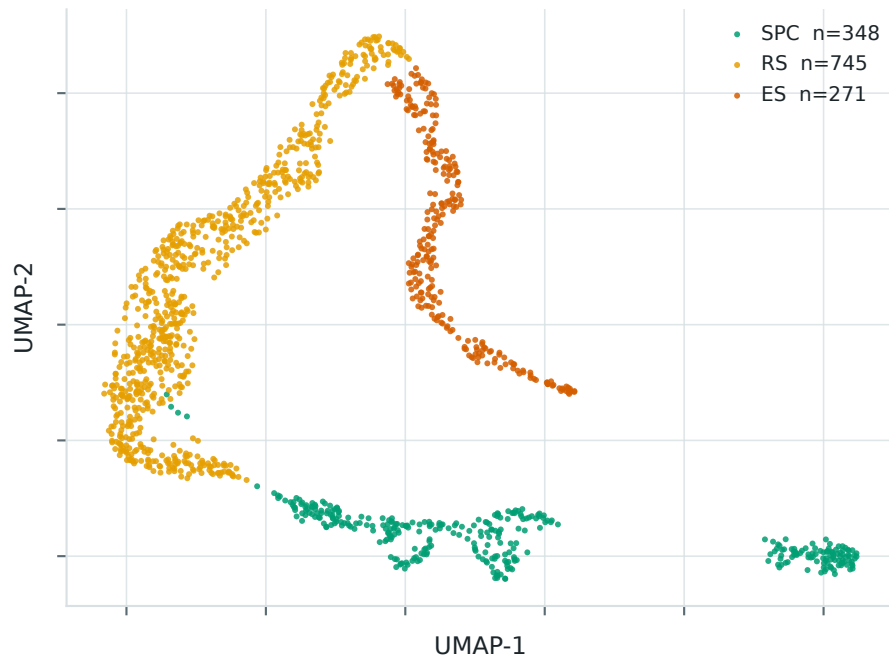
GSE104556 adult mouse testis, two mice analysed independently | PeakATail 0.2.0 count matrix re-keyed by pas_id (shipped matrix rows are mis-indexed; see caveats) | cells restricted to the 1,294 / 1,364 STARsolo-called cells

(a1) mouse1: stage from the PAS matrix alone



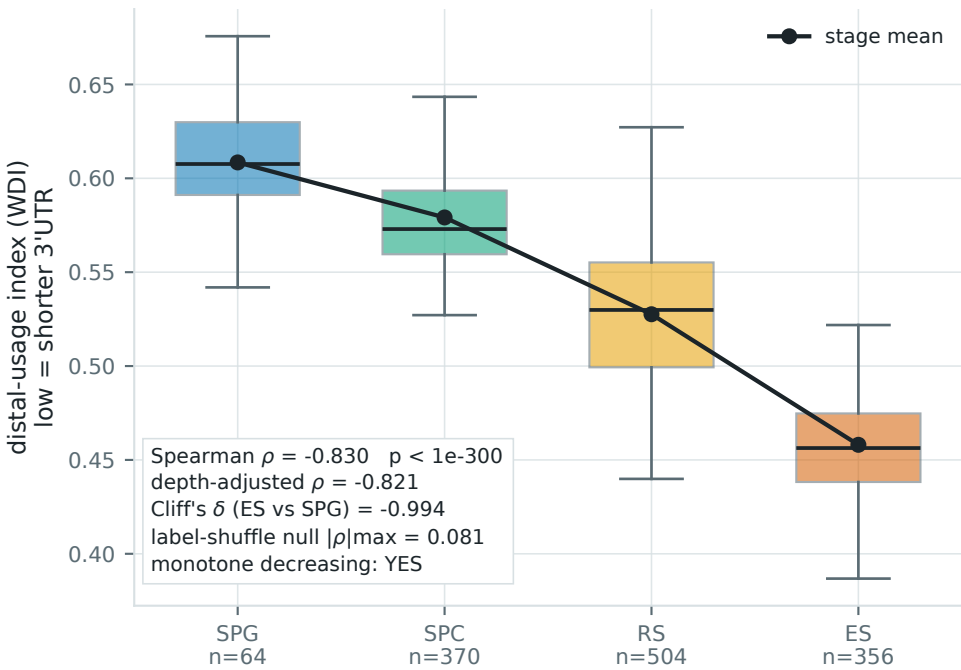
agreement with orthogonal STARsolo-GEX labels: 93.9%
PAS-only clustering merges the rare spermatogonia into SPC,
so panels b use the 4-stage STARsolo-GEX labels

(a2) mouse2: stage from the PAS matrix alone

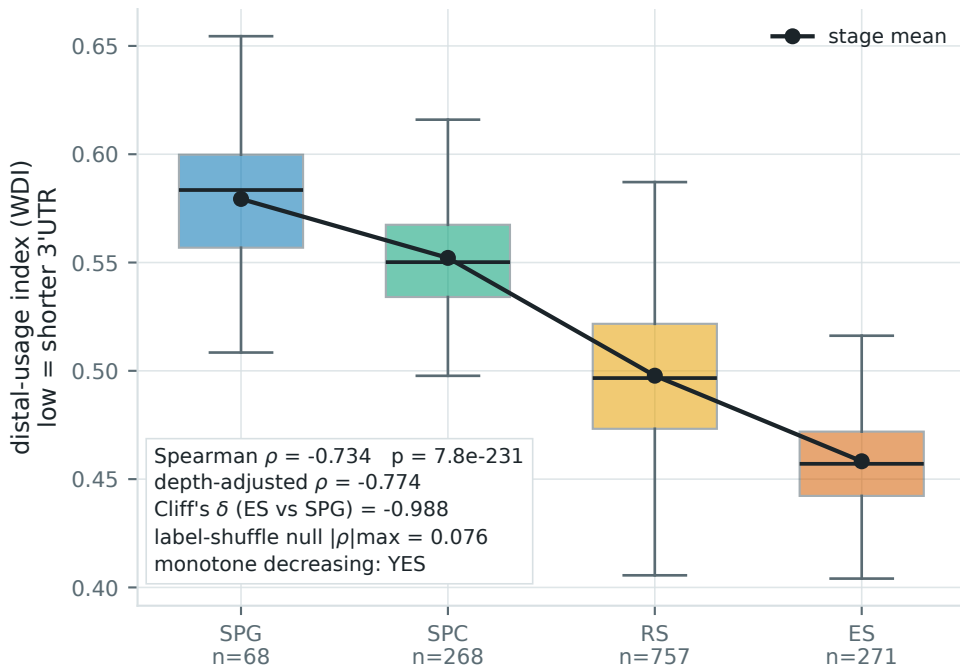


agreement with orthogonal STARsolo-GEX labels: 93.8%
PAS-only clustering merges the rare spermatogonia into SPC,
so panels b use the 4-stage STARsolo-GEX labels

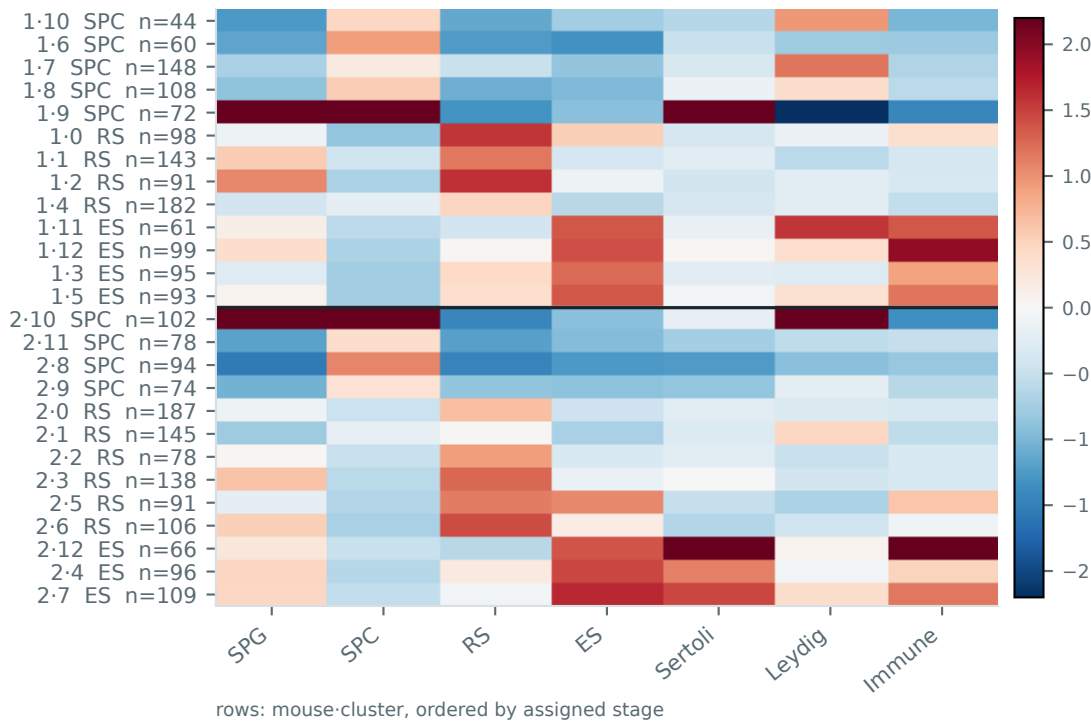
(b1) mouse1: 3'UTR length index across ordered stages (GEX labels)



(b2) mouse2: 3'UTR length index across ordered stages (GEX labels)

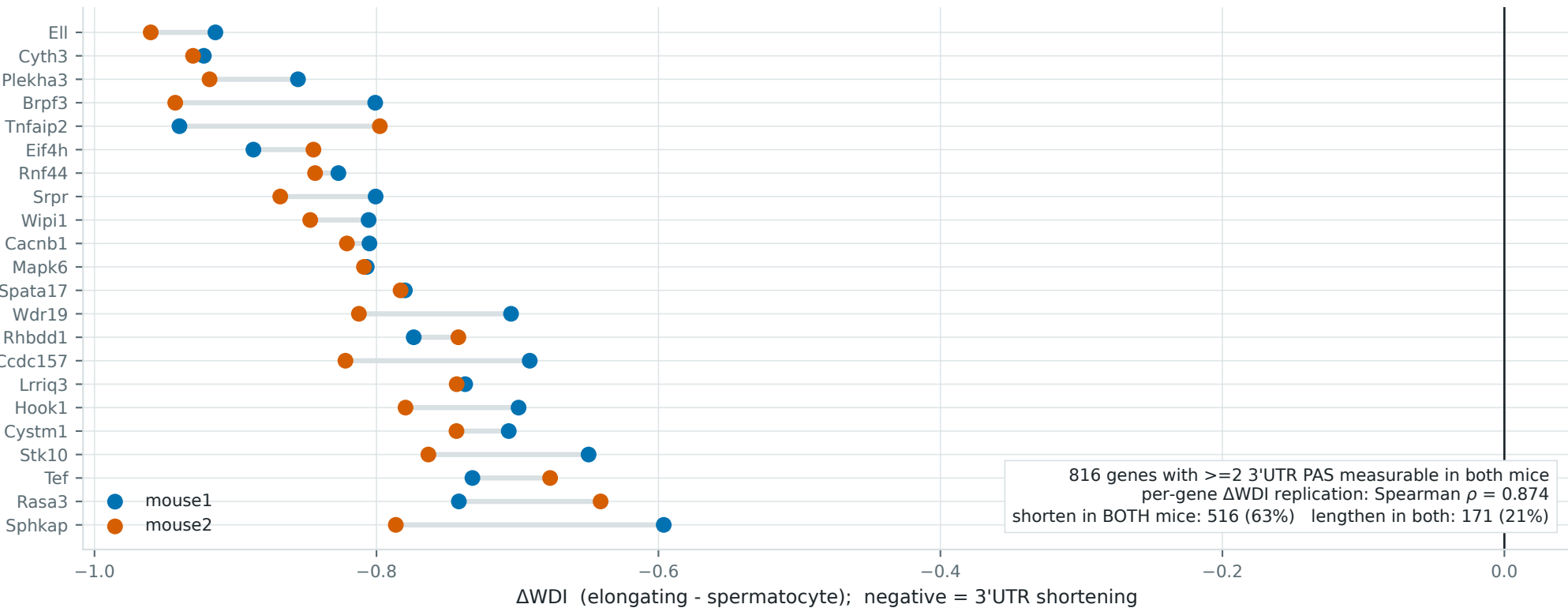


(d) marker-panel score per PAS-derived cluster (audit)

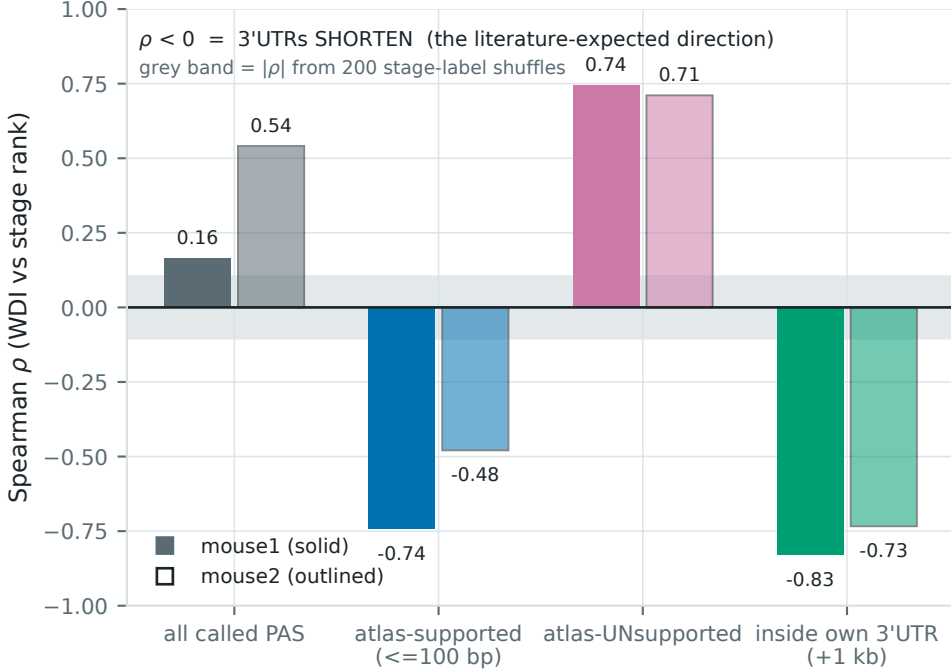


rows: mouse-cluster, ordered by assigned stage

(e) top shortening genes, replicated in both mice [3'UTR-restricted PAS]



(c) the trend lives ONLY in atlas-/3'UTR-supported PAS



CAVEATS -- read before quoting any number here

- The shipped matrix could not be used. ema/matrixfilter.py make_dataframe() returns a full-height matrix but a compacted pas_id list; annotate() then indexes rows by list POSITION, so every PAS carries another PAS's counts (drift 0->1,539 mouse1, 0->1,208 mouse2). Per-PAS counts vs samtools depth over the same interval: Spearman 0.058 as shipped -> 0.730 re-keyed. Everything above uses the re-keyed matrix.
- clusters.h5ad .X is TF-IDF, not counts, so 'ema switch length' as documented computes PDUI on IDF-weighted values. And its classic/per_gene path is strand-blind (proximal/distal swapped for every minus-strand gene; unit-tested). The index used here is computed directly from raw counts, strand-aware.
- Stages are MARKER-INFERRED, not sorted. The PAS-derived labels agree with orthogonal STARsolo-GEX labels at 93.9% / 93.8%, and every trend is reported under both labellings. Spermatogonia are rare (64 / 68 cells) and the PAS-only clustering merges them into spermatocytes.
- THE RED FLAG. With all called PAS the trend is absent or REVERSED; only 37% of PeakATail PAS are atlas-supported and 36% sit in an annotated 3'UTR. The atlas-UNsupported peaks carry a strong opposite-direction signal in both mice. They are modestly enriched for an internal-priming signature (A-run>=6 in +10...+40 nt: 5.4% vs 3.6%) and depleted of a canonical hexamer (4.2% vs 9.1%) -- suggestive, not conclusive, because PeakATail peaks are wide intervals rather than point cleavage sites.
- Depth is a confounder (spermatocytes carry ~2x the UMIs of elongating spermatids). Trends survive depth-adjusted partial correlation and binomial thinning to 6,000 UMI/cell.
- PeakATail's CB filter keeps 9,564 / 9,074 barcodes vs STARsolo's 1,294 / 1,364 cells; the extra ~86% are ambient droplets and are